

ANY RODRIGUES

Molecular Bioengineer | Neural Electrophysiology & HD-MEA

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Research Profile

Molecular bioengineer with hands-on experience in neural electrophysiology, high-density microelectrode arrays (HD-MEAs), ex vivo neural tissue, and in vitro neuronal models. Experienced in developing and optimizing electrophysiological experiments, extracellular neural recording and stimulation, tissue preparation, microscopy, and neural activity analysis. Seeking to apply this background to brain organoids, closed-loop electrophysiology, neural plasticity, and Organoid Intelligence.

Research Experience

Center for Regenerative Therapies Dresden (CRTD), TU Dresden

Dresden, Germany

Master's Researcher – Neural Electrophysiology

Feb 2024 – Aug 2024

Master's thesis: Establishing CMOS-MEA Recordings in Adult Zebrafish Retina

- Established and optimized electrophysiological recording protocols for ex vivo adult zebrafish retina using high-density CMOS-MEA platforms.
- Performed retinal tissue preparation and optimized tissue-electrode contact, perfusion, and recording conditions to improve signal quality and stability.
- Investigated spontaneous and light-evoked retinal ganglion cell activity under different visual stimulation paradigms.
- Characterized ON, OFF, and ON-OFF response profiles and investigated temporal responses to different light stimulation frequencies and light-dark transitions.
- Performed pharmacological experiments to investigate changes in retinal network activity.
- Processed and analysed extracellular neural recordings, including spike activity and network-level response characterization.
- Conducted extensive literature review and scientific evidence evaluation to support experimental design, interpretation, and contextualization of results.
- Worked independently while collaborating within an interdisciplinary neuroscience and bioengineering research environment.

German Center for Neurodegenerative Diseases (DZNE)

Dresden, Germany

Research Intern

Sep 2023 – Feb 2024

- Performed extracellular electrophysiological recordings using Axion Maestro 6-well and 24-well MEA systems and 3Brain high-density MEA platforms.
- Worked with human and rat olfactory bulb-olfactory epithelium co-culture models to investigate neural activity and network behaviour.
- Contributed to experimental workflows involving cultured neural systems and chemical stimulation.
- Conducted high-resolution microscopy imaging and biological sample characterization.
- Analysed neural activity data within a neurodegeneration-focused research environment.

Industry Experience

MaxWell Biosystems

Zürich, Switzerland

Account Manager Intern

May 2025 – Apr 2026

- Worked within a neurotechnology company developing and commercializing HD-MEA platforms for functional characterization of neuronal cultures and neural tissue.
- Gained close exposure to emerging applications of HD-MEA technology in brain-organoid electrophysiology and Organoid Intelligence.
- Developed a strong understanding of experimental workflows, neural recording technologies, and technical challenges surrounding high-density electrophysiology through interaction with researchers using advanced neural-interface platforms.
- Collaborated with scientific, technical, and commercial teams and translated technical and scientific information into structured insights.

Education

University of Porto	Porto, Portugal
MSc in Molecular Bioengineering	2021 – 2024

Thesis: *Establishing CMOS-MEA Recordings in Adult Zebrafish Retina*

Thesis research performed at the Center for Regenerative Therapies Dresden (CRTD), TU Dresden.

University of Porto	Porto, Portugal
BSc in Bioengineering	2018 – 2021

Thesis: *Development of optical sensors using molecularly imprinted polymers for ferritin detection*

Technical Expertise

Electrophysiology & Neural Interfaces

HD-MEA; CMOS-MEA; Axion Maestro MEA; 3Brain HD-MEA platforms; extracellular neural recording; extracellular stimulation; neural network activity characterization; electrophysiological data analysis.

Experimental Models & Tissue Work

Ex vivo adult zebrafish retina; retinal dissection and tissue preparation; primary cell culture; human and rat olfactory bulb–olfactory epithelium co-cultures; biological sample preparation.

Microscopy & Imaging

Confocal microscopy; high-resolution microscopy; Z-stack imaging; biological sample characterization; image acquisition and analysis.

Data Analysis & Programming

Python for neural data analysis, electrophysiology data processing, spike-train analysis, and data visualization; Microsoft Excel.

Research & Experimental Practice

Experimental design; protocol development and optimization; independent project execution; scientific literature review; scientific reporting; interdisciplinary collaboration; experimental troubleshooting and problem solving.

Animal Research

Zebrafish handling and experimental procedures; FELASA-accredited laboratory animal training.

Certifications & Specialized Training

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- **FELASA-Accredited Laboratory Animal Training** – training in laboratory animal science and experimental animal research.
 - **Zebrafish Handling & Experimental Procedures** – practical training in zebrafish handling and experimental procedures, Center for Regenerative Therapies Dresden (CRTD), TU Dresden.
 - **Introduction to Python and Machine Learning for the Biosciences** – i3S – Institute for Research and Innovation in Health, Porto, Portugal.

Research Interests

Organoid Intelligence · Brain Organoids · HD-MEA Electrophysiology · Closed-Loop Stimulation
Neural Plasticity · Biological Computing · Neural Interfaces · Functional Neural Networks

Languages

Portuguese – Native · English – Fluent · German – Beginner · French – Basic

Research Direction

My research interests have developed from experimental neural electrophysiology and high-density recording of living neural systems toward brain-organoid electrophysiology and biological computation. I am particularly interested in experimental systems that combine organoid culture, HD-MEA recording, stimulation, and closed-loop interaction to investigate how living neural networks process information, and potentially adapt and learn.